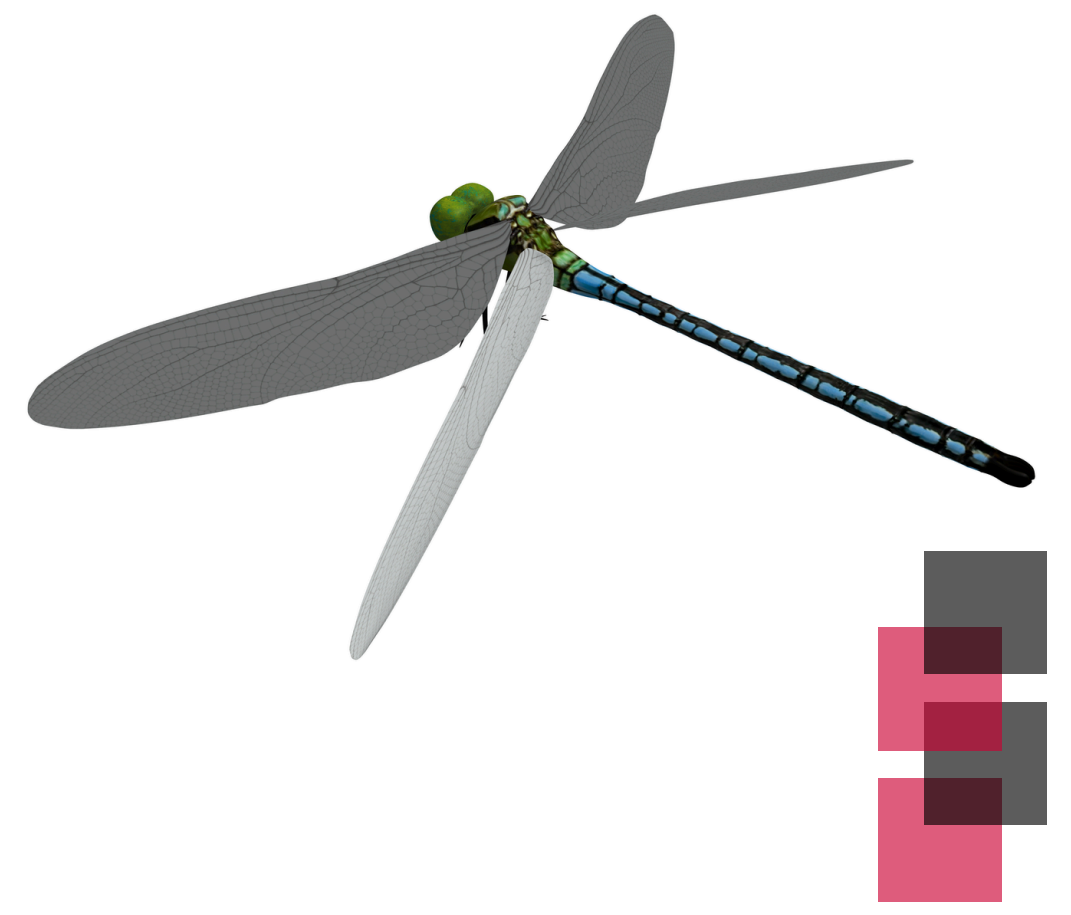


Flying Insects and Flapping Robots

Visualizing Wing Rotation and Fluid Flow in 3 Dimensions

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Introduction

This research introduces visualization tools to understand and simulate insect flight dynamics, aiming to advance flight technologies and inspire insect-like robots. Current fluid simulation methods, like Navier-Stokes solutions, are computationally intensive. Our project uses the efficient Lagrangian-based Vortex Filament Method (VFM) with an elegant quaternion-based 3D rotation model. We study velocity fields, vortex strengths, forces, and flight paths by modeling different wing flapping patterns inspired by real insects. Our VFM approach enables near real-time visualization of fluid flow and insect flight, allowing for use in optimizing flight technologies and robotic applications.

Methods and Discussion

Rotations

Rotations in 3D are inherently challenging because they are non-commutative—changing the order in which rotations are applied changes the final orientation.

Quaternions, a four-component extension of complex numbers, elegantly represent rotations and offer several advantages:

- Combine rotations easily: Multiplying quaternions produces a single equivalent rotation, avoiding Euler angle complexities.
- Efficient interpolation: SLERP provides smooth, stable orientation interpolation along the 4 dimensional constraint manifold of the state space. It is simpler and faster than with Euler angles or matrices.
- Compact & versatile: Only 4 entries (vs. 9 for matrices), easy to invert, and well-suited for coordinate transforms, rotations, and reflections.

$$q = a + bi + cj + dk$$

$$i^2 = j^2 = k^2 = ijk = -1$$

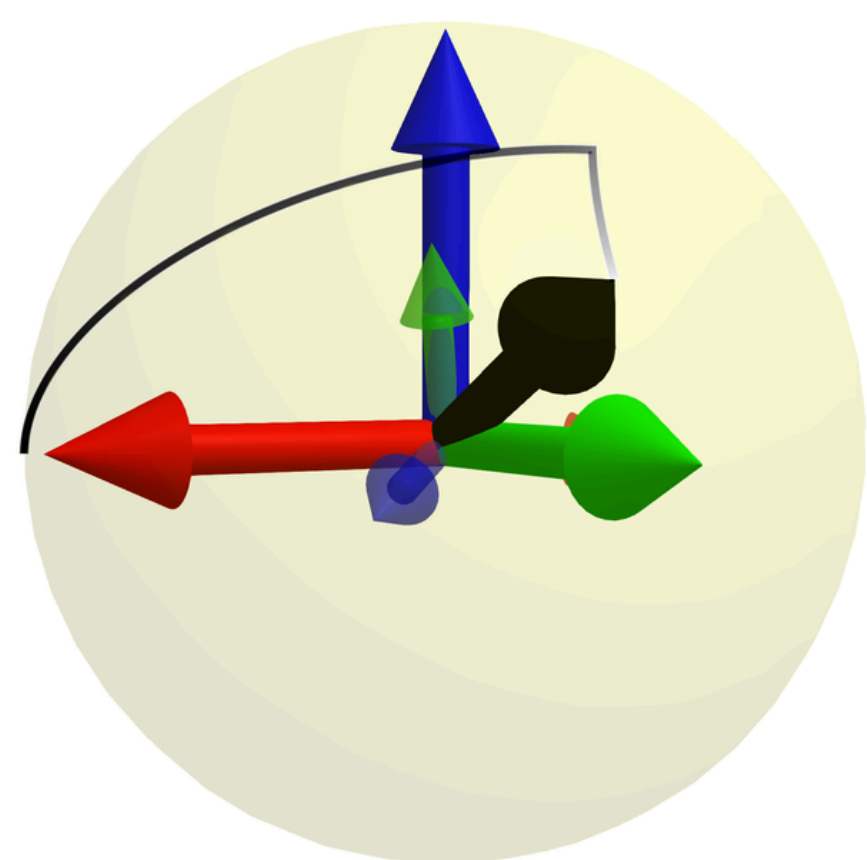


Fig. 1. Demonstration of Spherical Linear Interpolation (SLERP) to smoothly transition between rotational states along the constraint manifold.

Fluid Dynamics

Definitions

A **vortex** is a swirling region of fluid driven by **vorticity**, indicating rotational motion. **Circulation** measures the rotational strength around a closed loop. A **vortex filament** models concentrated vorticity along a thin curve, while a **vortex loop** is a closed filament, used because open-ended filaments are unstable in inviscid flow.

We model airflow around insect wings using a grid of vortex loops aligned with the wing surface, which are later shed into the surrounding air.

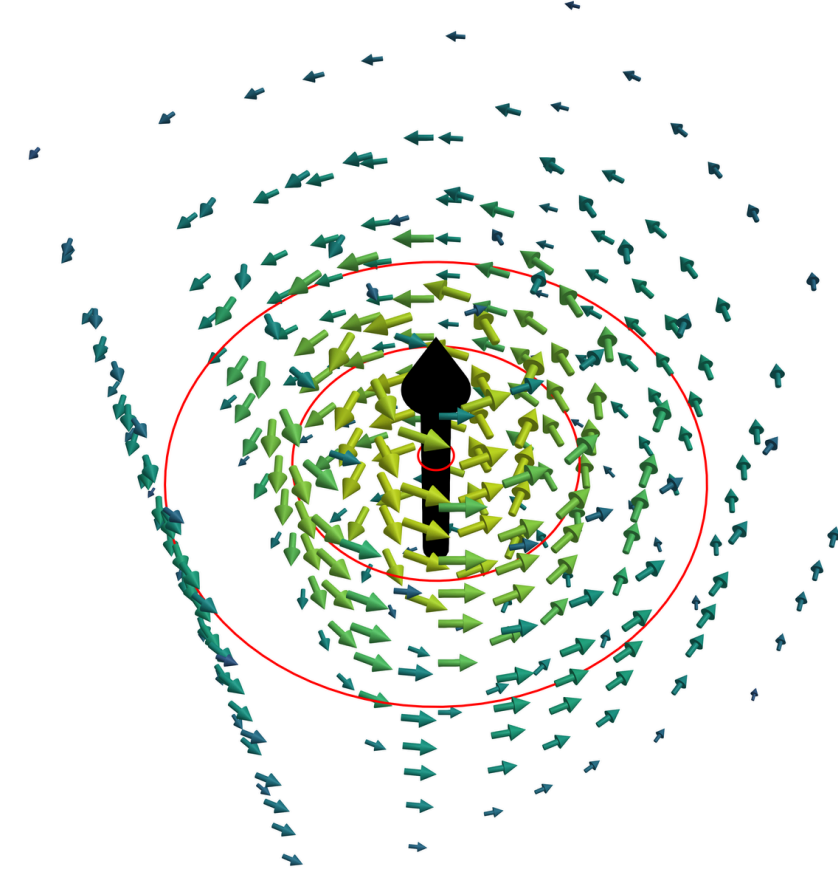


Fig. 2. Visualization of the velocity field (arrows color-coded Blue to Yellow) due to a Vortex Filament (Black arrow). The red contour lines are added for improved visuals. Airflow can be seen circulating around the vortex line.

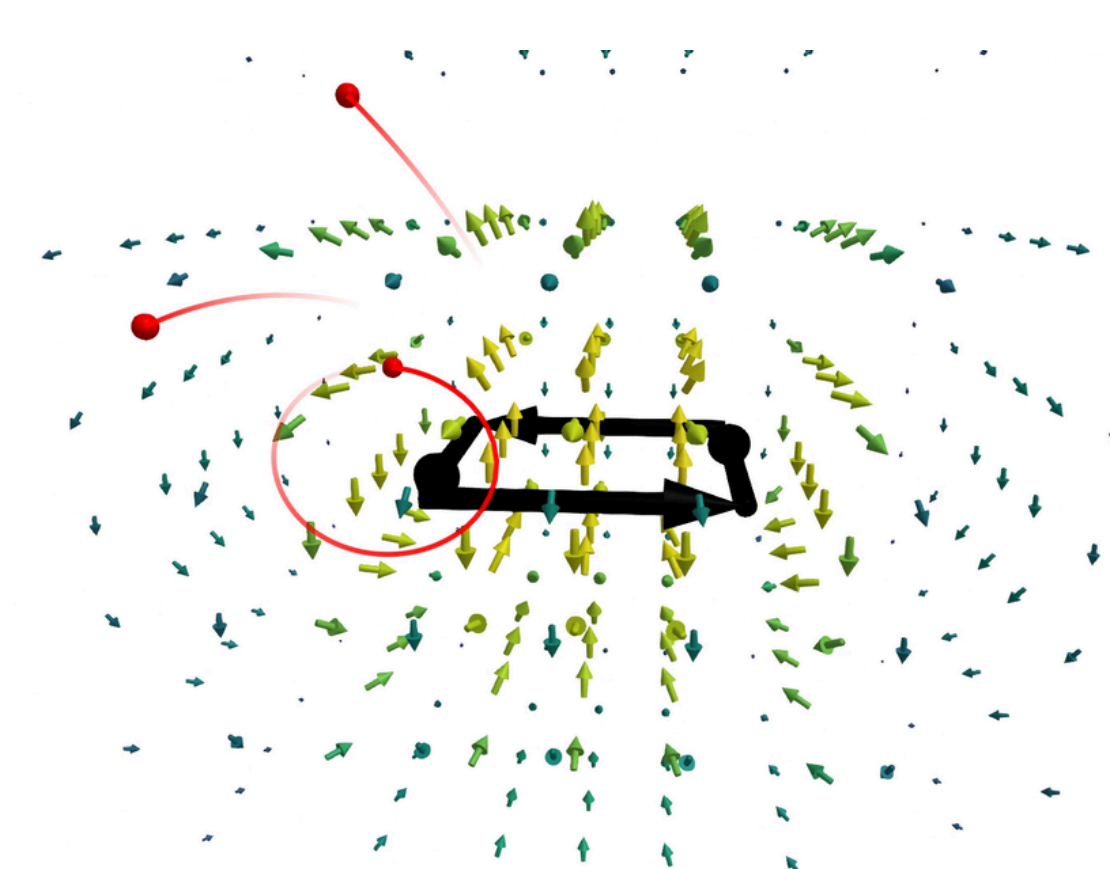


Fig. 3 Visualization of the velocity field (arrows color-coded Blue to Yellow) due to a Vortex Loop (Black arrows) composed of 4 Vortex Lines. Fluid particles' (Red) motion has been simulated using Runge-Kutta 4 as a red trail.

Simulation

By combining the ideas of rotations and fluid flow, we bring an insect flight simulation to life.

Developed using the Julia programming language, a robust scientific computing platform, it employs:

- Makie.jl for visualizations.
- Quaternions.jl for rotational calculations.

Some computational routines are adapted from existing references, while others are original, derived through analytical methods and iterative refinement. The simulation animates wing movements and visualizes fluid flow for arbitrary flapping patterns, defined by angular velocity functions for roll, pitch, and yaw.

Inspired by dragonfly flight, a demonstration models a dragonfly, with independently controlled amplitudes, frequencies, and phases for roll and pitch of front and hind wings.

Following the same steps, any insect flight can be modelled and simulated.

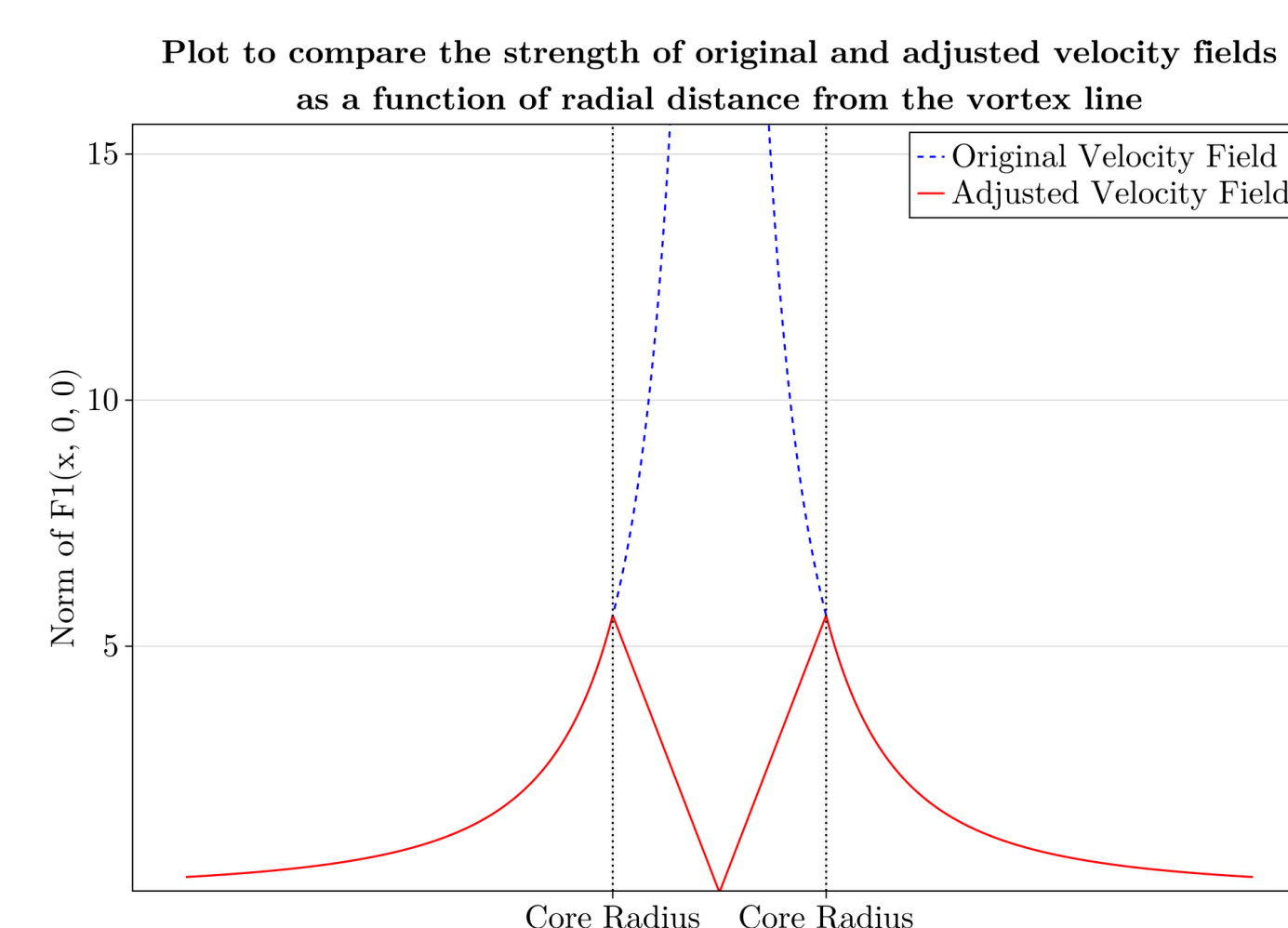


Fig. 4. Plot comparing the velocity field strengths as calculated by Biot-Savart law and modified Biot-Savart law.

Results and Conclusions

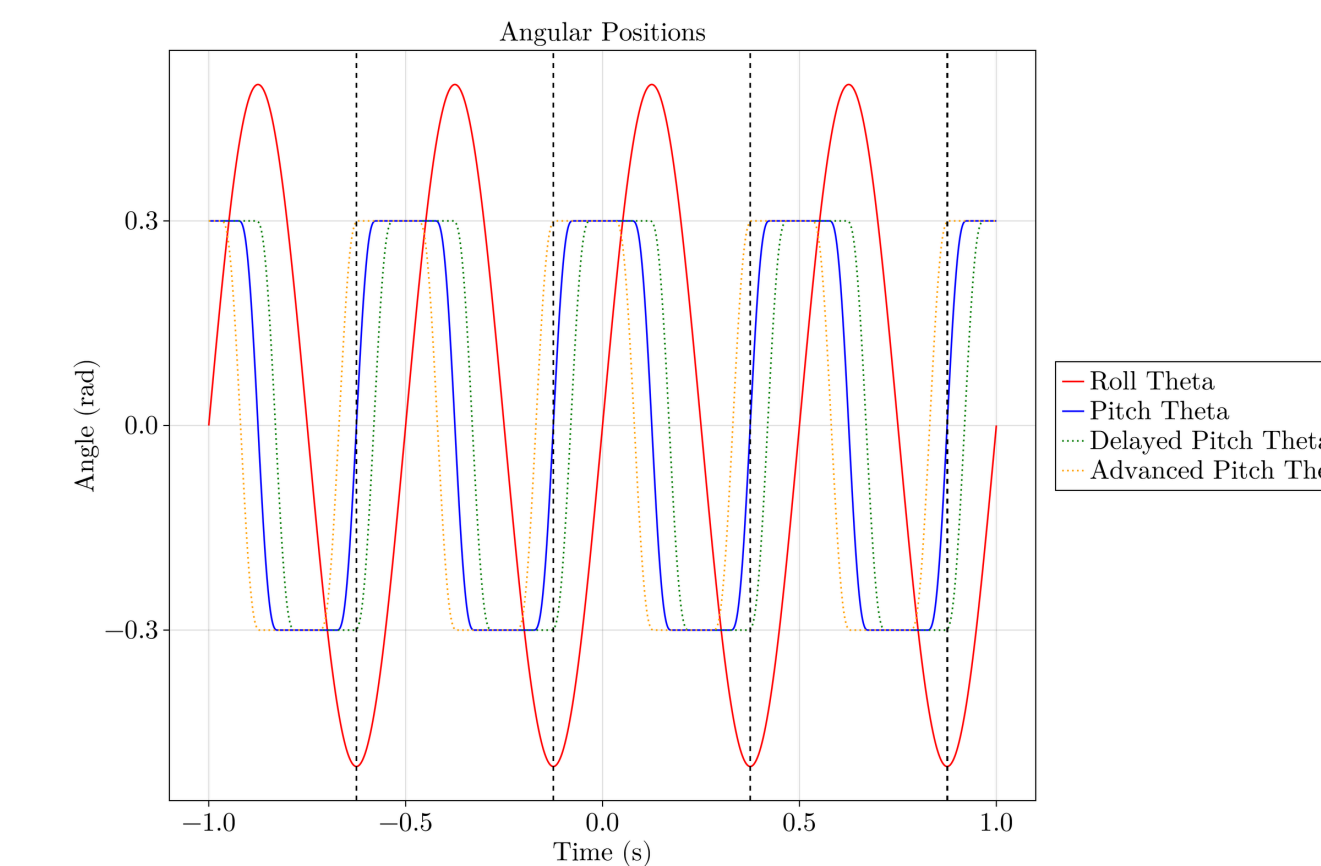


Fig. 5. Plot for the front wing flapping angles over time. Roll (flapping around the X-axis) is sinusoidal, and pitch (attack angle) is a symmetric smoothstep function. Advanced and delayed phases, shown in dotted lines, assist in ascent and turning maneuvers.

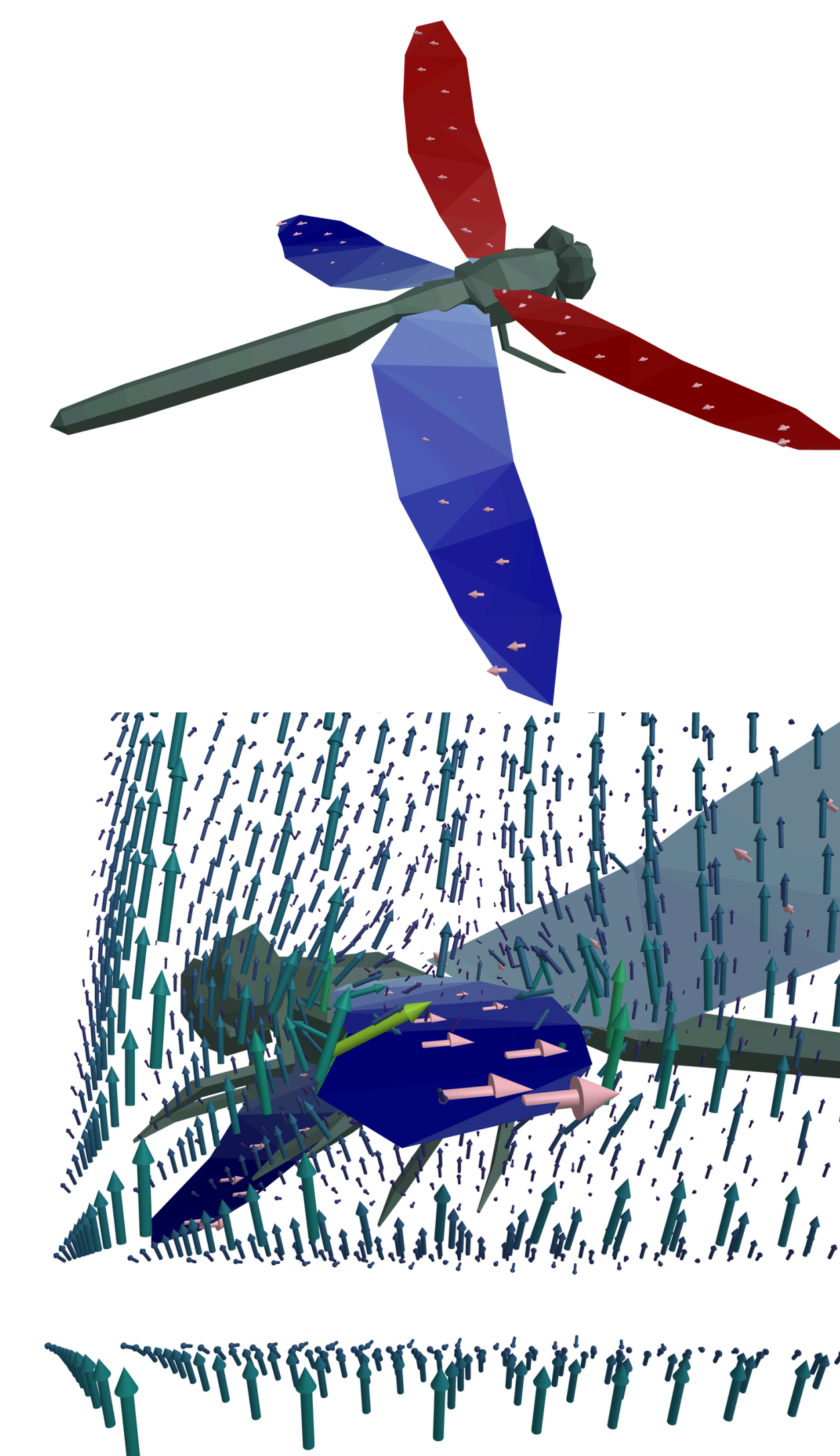


Fig. 6. Frame from the dragonfly simulation showing a low-polygon wing mesh tiled by vortex loops along mesh face edges (triangular). Wing motion (from Fig. 5) defines circulation strengths via the non-penetration condition, visualized by face shading (red = positive Γ , blue = negative Γ , darker = higher magnitude). Pink arrows show relative fluid flow at loop centroids, lying in the wing plane as expected.

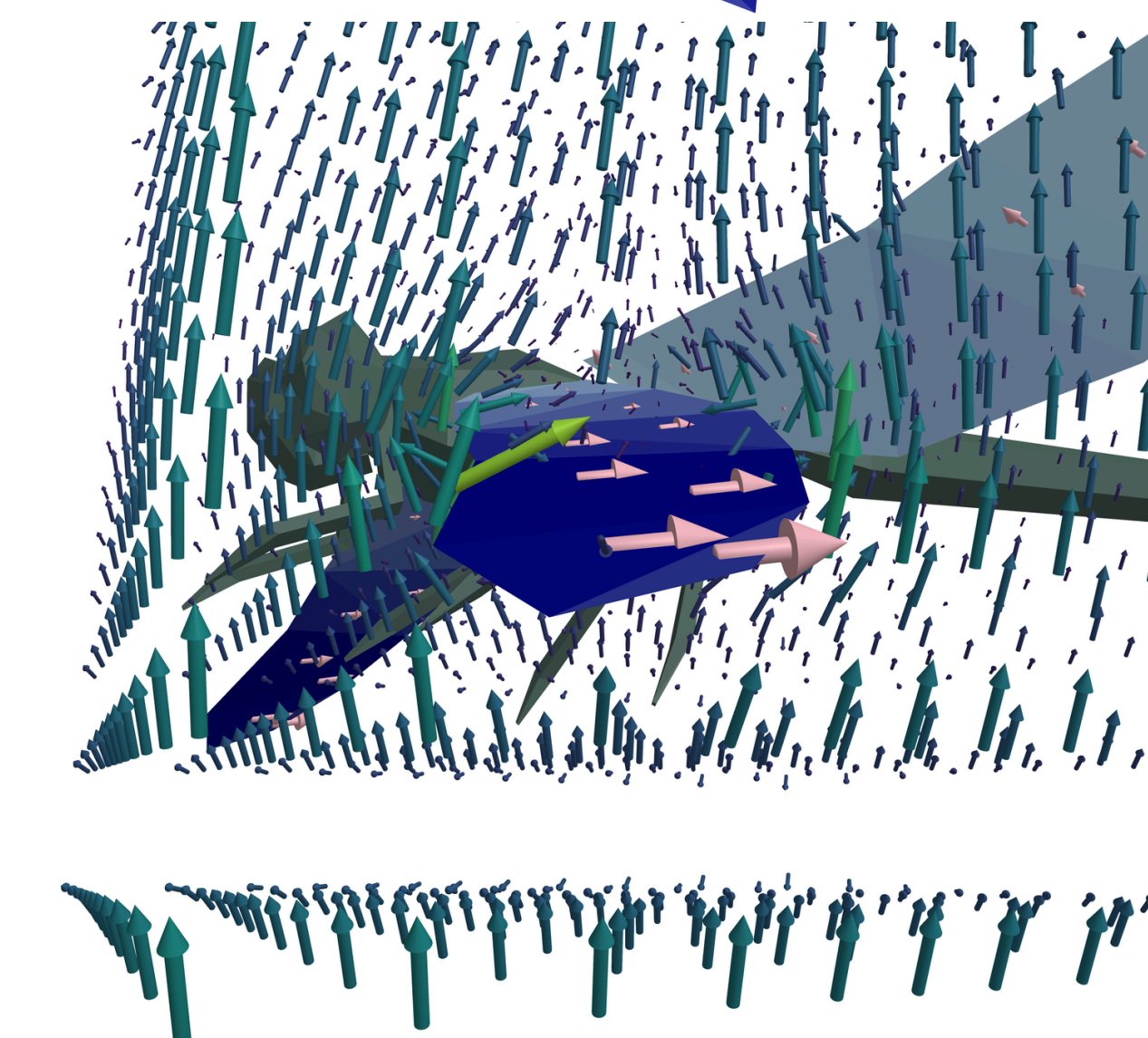


Fig. 7. Fluid flow around the dragonfly at $t=0.1$ s for the wing motion defined in Fig. 5. The dragonfly body mesh matches Fig. 6 but at a different time stamp. The view focuses on the airflow near the tip of the front left wing. Pink arrows show relative airflow at loop centroids (as in Fig. 6), and flow vector colors (blue to yellow, viridis colormap as in Figs. 2–3) indicate airflow speed elsewhere in space.

The simulation successfully handles arbitrary rotation parameters, effectively modeling a variety of wing flapping mechanisms with precision.

In the particular case of the dragonfly simulation, we accurately simulate a 3D dragonfly model with correct rotations, observing that the airflow near the base of the wings is minimal as expected. The circulation strength (Gamma) is also lowest near the wing base and reaches its highest values at the tips, with Gamma values changing sign (indicated by red and blue colors) as the direction of wing flapping shifts.

Our Vortex Filament Method reliably models the bulk airflow patterns, where fluid particles tend to swirl energetically around the edges and move slowly in a streamlined manner further away from the wing edges.

References

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Future Directions

We aim to extend the simulation to model vortex shedding and wake vortices, calculating insect flight paths based on the overall airflow. We hope to benchmark it against existing models and develop a controllable simulation that adjusts wing flapping dynamically to achieve precise flight movements. Finally, we also aim to implement this in a mechanical flying robot model.

Scan the QR code for videos, simulation testing and project updates!

